

# Fundamentals of Cryptography and Information Security Study

Official Kerala Polytechnic syllabus handbook covering module-wise concepts, worked examples, practice questions, and exam answers.

## Course Code

3283

## Credits

4

## Periods

5/week

## Course Details

**Title:** Fundamentals of Cryptography and Information Security Study

**Department:** Common

**Revision:** REV\_Fundamentals of Cryptography and Information Security Study

**Pattern:** Theory / Practical

Complies with official SITTTR guidelines with Malayalam support notes and worked exercises.

Curriculum integrity

## Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 3283 — Fundamentals of Cryptography and Information Security Study**. Use the official SITTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

[Official SITTTTR Revision 2026 syllabus reference](#). This page is a study handbook, not a substitute for the latest official notification or examination instruction.

**Malayalam support:** ു ു൬൬൬൬൬൬ ു൬൬൬൬൬൬ ു൬൬൬൬൬൬ English technical terms- ു൬൬൬൬൬൬ ു൬൬൬൬൬൬. ു൬ ു൬൬൬൬൬ ു൬൬൬൬൬൬ ു൬൬൬൬൬൬, formula, diagram, units, definitions ു൬൬൬ ു൬൬൬൬൬൬൬ ു൬൬൬൬൬൬൬ ു൬൬൬൬.

**Common mistakes and precaution:** Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

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#### OBJECTIVES

## Course Objectives and Outcomes

### Objectives

1. Provide deep technical understanding of Fundamentals of Cryptography and Information Security Study.
2. Develop practical competencies aligned with industry standards.
3. Prepare students for board examinations.

## CO-PO Mapping

CO1 → PO1: 3. CO2 → PO2: 3. CO3 → PO3: 3. CO4 → PO3: 3.

### MODULE 1

## Core Principles

### Outcomes

Master foundational terminology and core definitions of Fundamentals of Cryptography and Information Security Study.

### Study

Detailed analysis of core principles and theoretical foundations.

**Malayalam Support Note:** ക്രിപ്റ്റോഗ്രഫിയിലെ അടിസ്ഥാനപദങ്ങളും വിവരസുരക്ഷയുടെ ആവശ്യകതയും പഠിക്കുക. ക്രിപ്റ്റോഗ്രഫിയിലെ അടിസ്ഥാനപദങ്ങളും വിവരസുരക്ഷയുടെ ആവശ്യകതയും പഠിക്കുക.

### MODULE 2

## Applied Methods

### Outcomes

Apply standard calculation techniques and operational procedures.

## Study

Numerical problem-solving techniques and practical workflows.

**Malayalam Support Note:** ഐക്യരൂപം ഉപയോഗിച്ച് സംയോജിതമായ സിസ്റ്റം വിശ്ലേഷണം നടത്തുക. ഐക്യരൂപം ഉപയോഗിച്ച് സംയോജിതമായ സിസ്റ്റം വിശ്ലേഷണം നടത്തുക.

### MODULE 3

## Advanced Integration

### Outcomes

Evaluate complex systems and optimize performance.

## Study

Advanced system integration and troubleshooting methodologies.

**Malayalam Support Note:** സങ്കീർണ്ണ സിസ്റ്റം വിശ്ലേഷണം നടത്തുക. സങ്കീർണ്ണ സിസ്റ്റം വിശ്ലേഷണം നടത്തുക. ഐക്യരൂപം ഉപയോഗിച്ച് സംയോജിതമായ സിസ്റ്റം വിശ്ലേഷണം നടത്തുക.

### MODULE 4

## Synthesis & Case Studies

### Outcomes

Design comprehensive technical solutions and demonstrate mastery.

## Study

Real-world case studies and industry project design.

**Malayalam Support Note:** ഐക്യം നമ്മുടെ ഏകതയാണ്. ഐക്യം നമ്മുടെ ഏകതയാണ്. ഐക്യം നമ്മുടെ ഏകതയാണ്.

## MODEL PAPER

# Practice Model Question Paper

**Time: 3 Hours | Max Marks: 75**

### Part A (2 marks each)

1. Define core concepts of Fundamentals of Cryptography and Information Security Study.
2. Explain fundamental principles in Module 1.
3. Discuss calculation methods in Module 2.
4. Describe advanced integration in Module 3.
5. Summarize case studies in Module 4.

### Part B (5 marks each)

1. Detailed explanation of Module 1 concepts with examples.
2. Comprehensive analysis of Module 2 methods.
3. System integration and troubleshooting in Module 3.
4. Industry applications and design in Module 4.

## ANSWERS

## Full Answer Key

**Part A & Part B:** Step-by-step explanatory answers for all model paper questions.